

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
from google.colab import files

uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving netflix_titles.csv to netflix_titles.csv

```
df = pd.read_csv("netflix_titles.csv")

print(df.head())
```

	show_id	type	title	director	cast
0	s1	Movie	Netflix Title 1	NaN	Actor E, Actor F
1	s2	TV Show	Netflix Title 2	NaN	Actor E, Actor F
2	s3	TV Show	Netflix Title 3	NaN	Actor A, Actor B
3	s4	Movie	Netflix Title 4	NaN	Actor C, Actor D
4	s5	Movie	Netflix Title 5	Director B	NaN

	country	date_added	release_year	rating	duration
0	Spain	September 5, 2018	2012	NR	109 min
1	India	October 30, 2020	2011	TV-14	4 Seasons
2	Canada	May 29, 2021	2021	PG-13	3 Seasons
3	United Kingdom	February 24, 2021	2011	G	126 min
4	France	February 10, 2018	2021	TV-MA	63 min

	listed_in	description
0	International TV Shows	Description of Netflix Title 1
1	International Movies	Description of Netflix Title 2
2	Comedies	Description of Netflix Title 3
3	Thrillers	Description of Netflix Title 4
4	International Movies	Description of Netflix Title 5

```
print("Shape of dataset:", df.shape)
print("\nColumn names:\n", df.columns.tolist())
```

Shape of dataset: (8807, 12)

Column names:

['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description']

```
print("\nData types & non-null info:")
df.info()
```

```
Data types & non-null info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        5271 non-null   object
4   cast            5218 non-null   object
5   country         8807 non-null   object
6   date_added      8807 non-null   object
7   release_year    8807 non-null   int64
8   rating          8807 non-null   object
9   duration        8807 non-null   object
10  listed_in       8807 non-null   object
11  description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
print("\nSummary statistics (categorical columns):")
print(df.describe(include="object"))
```

Summary statistics (categorical columns):

	show_id	type	title	director	cast \	
count	8807	8807	8807	5271	5218	
unique	8807	2	8807	3	3	
top	s8807	Movie	Netflix Title	8807	Director A, Actor B	
freq	1	6246	1	2147	2151	

	country	date_added	rating	duration	listed_in \
count	8807	8807	8807	8807	8807
unique	9	1362	8	126	10
top	South Korea	April 3, 2020	TV-MA	1 Seasons	Thrillers
freq	1066	15	2822	551	927

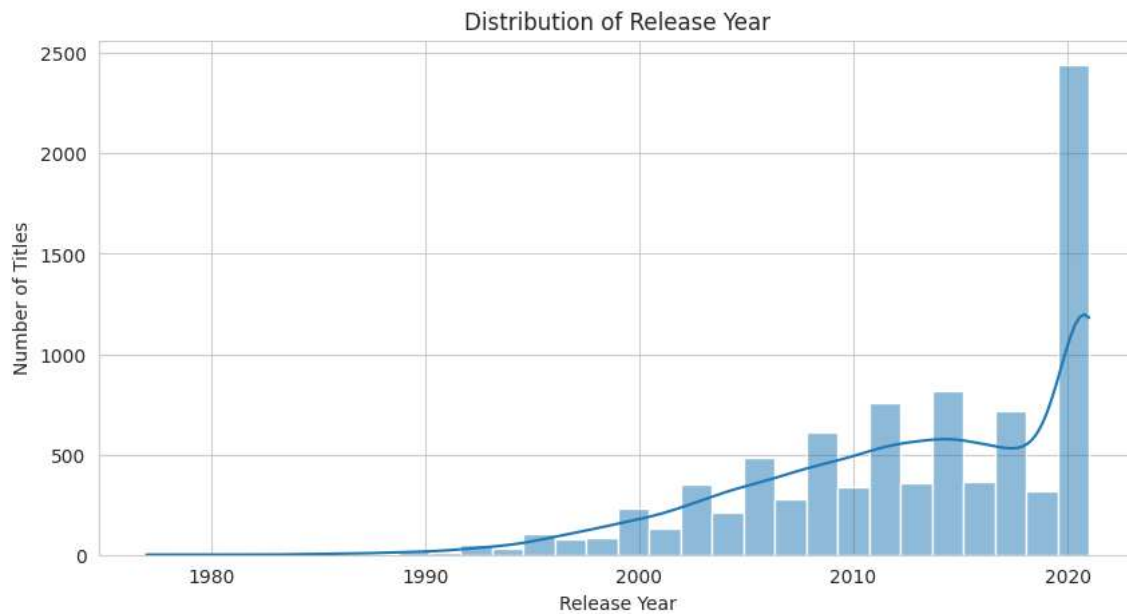
	description
count	8807
unique	8807
top	Description of Netflix Title
freq	1

```
plt.figure(figsize=(10,5))

sns.histplot(df["release_year"], kde=True, bins=30)

plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")

plt.show()
```

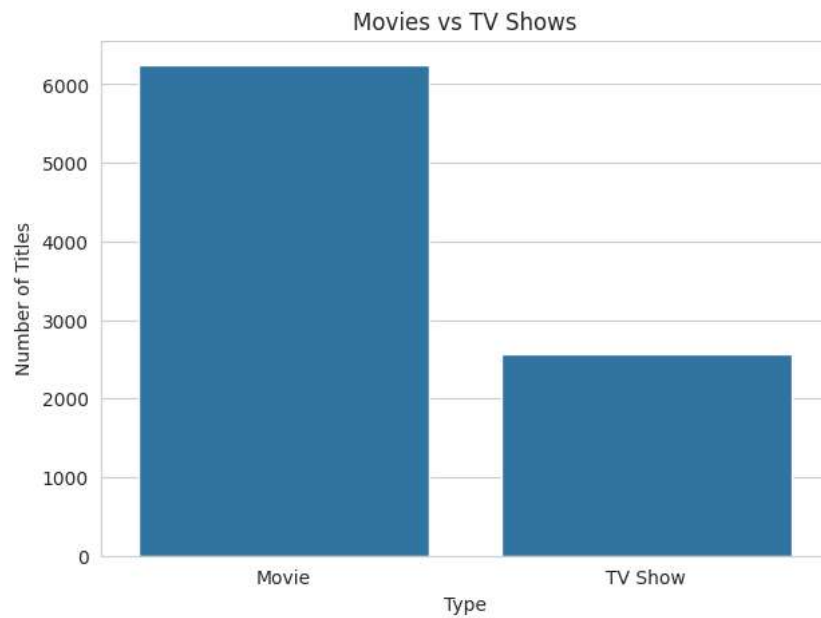


```
plt.figure(figsize=(7,5))

sns.countplot(data=df, x="type")

plt.title("Movies vs TV Shows")
plt.xlabel("Type")
plt.ylabel("Number of Titles")

plt.show()
```

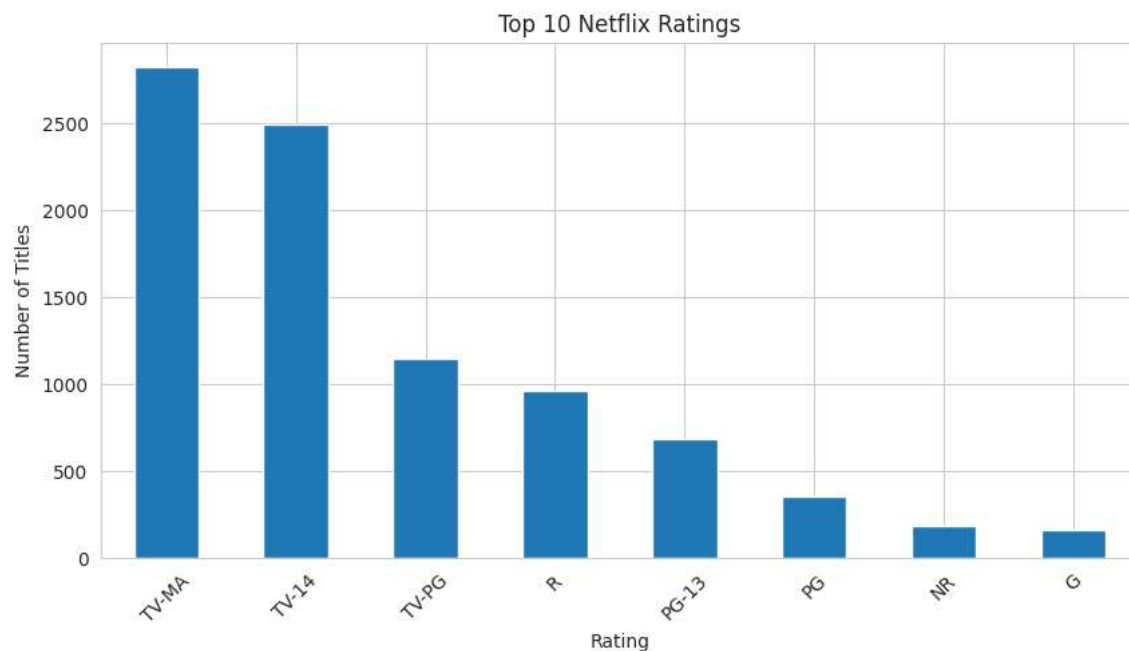


```
plt.figure(figsize=(10,5))

df["rating"].value_counts().head(10).plot(kind="bar")

plt.title("Top 10 Netflix Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

plt.show()
```

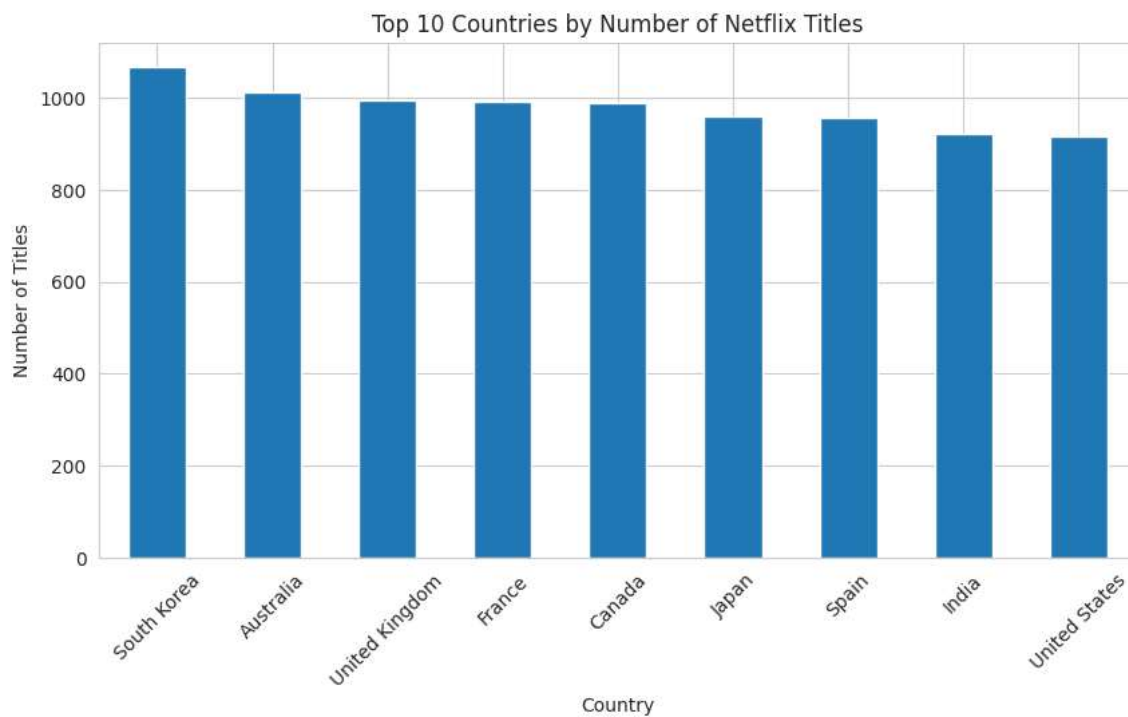


```
plt.figure(figsize=(10,5))

df["country"].value_counts().head(10).plot(kind="bar")

plt.title("Top 10 Countries by Number of Netflix Titles")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

plt.show()
```



```
corr = df.select_dtypes(include="number").corr()
```

```
print("Correlation matrix:\n")  
print(corr)
```

Correlation matrix:

```
          release_year  
release_year         1.0
```

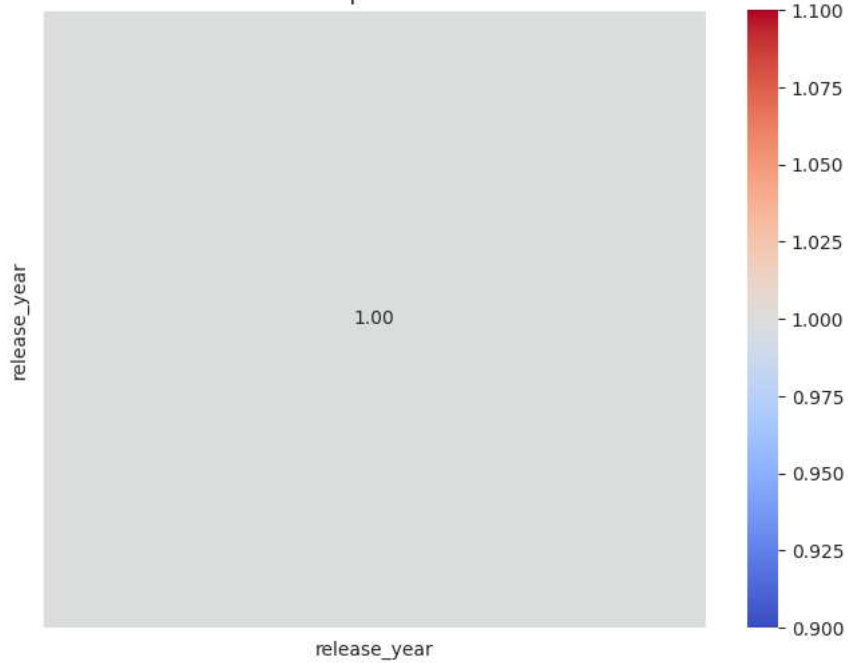
```
plt.figure(figsize=(8,6))
```

```
sns.heatmap(  
    corr,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f",  
    linewidths=0.5  
)
```

```
plt.title("Correlation Heatmap of Numerical Features")
```

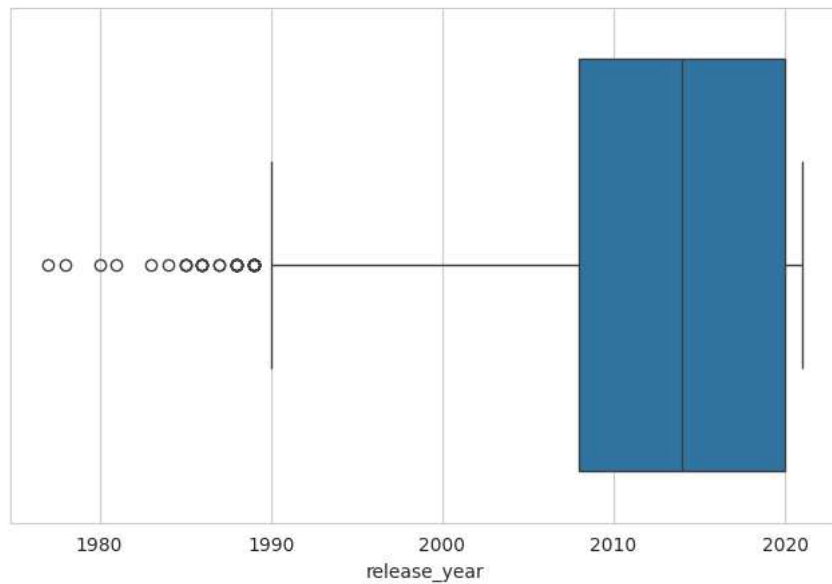
```
plt.show()
```

Correlation Heatmap of Numerical Features



```
plt.figure(figsize=(8,5))
sns.boxplot(x=df["release_year"])
plt.title("Box Plot of Release Year")
plt.show()
```

Box Plot of Release Year

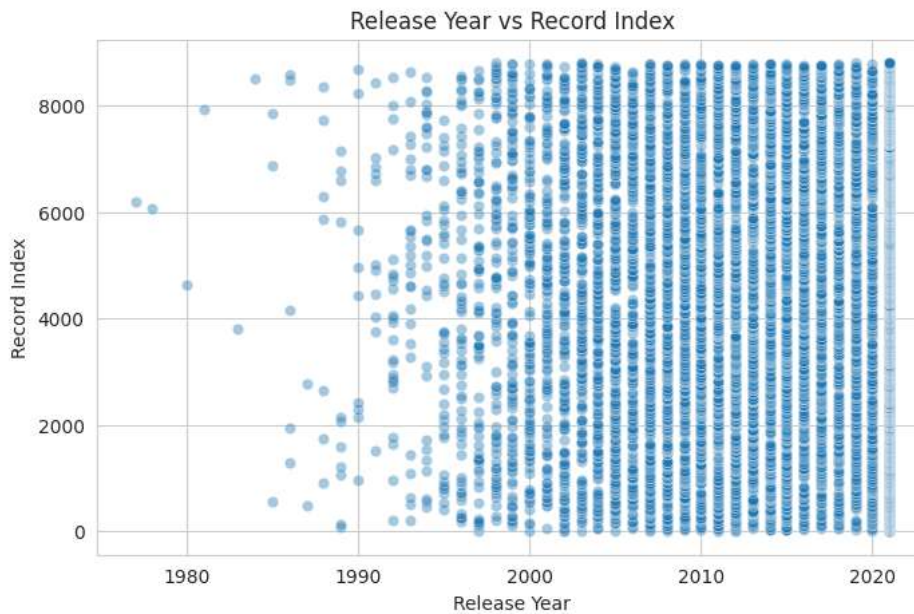


```
plt.figure(figsize=(8,5))

sns.scatterplot(
    data=df,
    x="release_year",
    y=df.index,
    alpha=0.4
)

plt.title("Release Year vs Record Index")
plt.xlabel("Release Year")
plt.ylabel("Record Index")
```

```
plt.show()
```

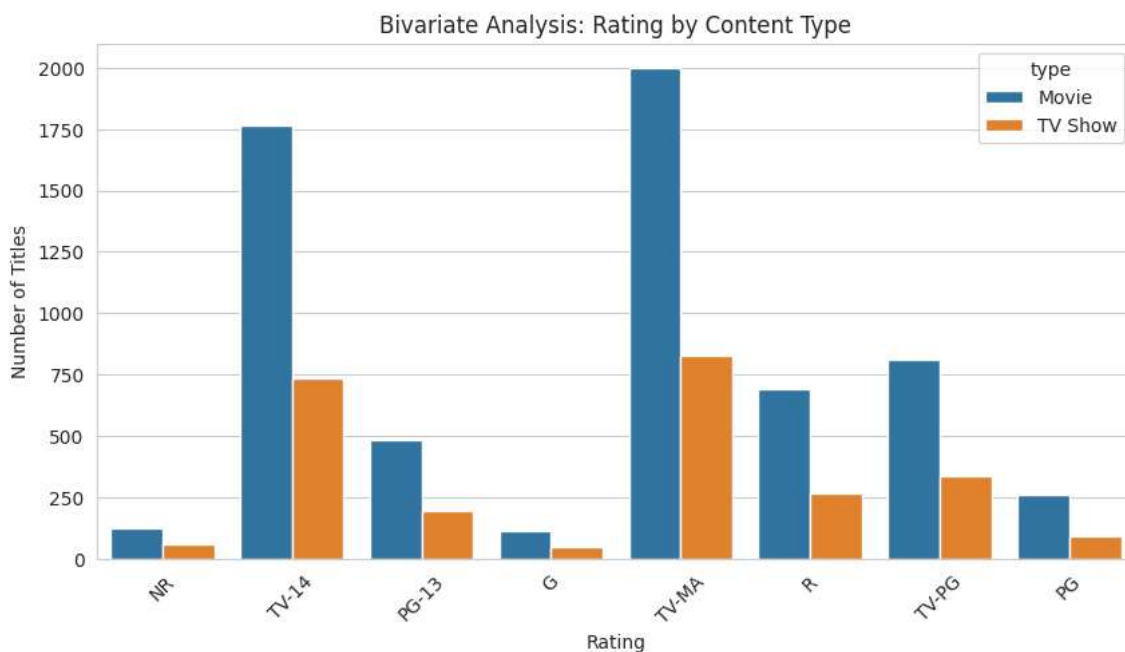


```
plt.figure(figsize=(10,5))

sns.countplot(
    data=df,
    x="rating",
    hue="type"
)

plt.title("Bivariate Analysis: Rating by Content Type")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

plt.show()
```



```
top_countries = df["country"].value_counts().head(5).index

temp = df[df["country"].isin(top_countries)]

plt.figure(figsize=(10,6))
```

```
sns.countplot(  
    data=temp,  
    x="country",  
    hue="type"  
)  
  
plt.title("Multivariate Analysis: Content Type by Country")  
plt.xlabel("Country")  
plt.ylabel("Number of Titles")  
plt.xticks(rotation=45)  
  
plt.show()
```

